

7. (a) (i) **Complete** the list showing the parts of the electromagnetic spectrum in order. [2]

gamma rays

X-rays

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visible light

infra-red

.....

radio waves

- (ii) State **one** property that is common to all electromagnetic waves. [1]

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- (b) Visible light has been used to accurately measure the distance between Earth and the Moon. In 1969, Neil Armstrong and Buzz Aldrin became the first humans to stand on the Moon. One of the jobs that they were required to do was to set up a reflector that faced Earth to allow light to be sent from the Earth, directed at it and be received back at Earth.

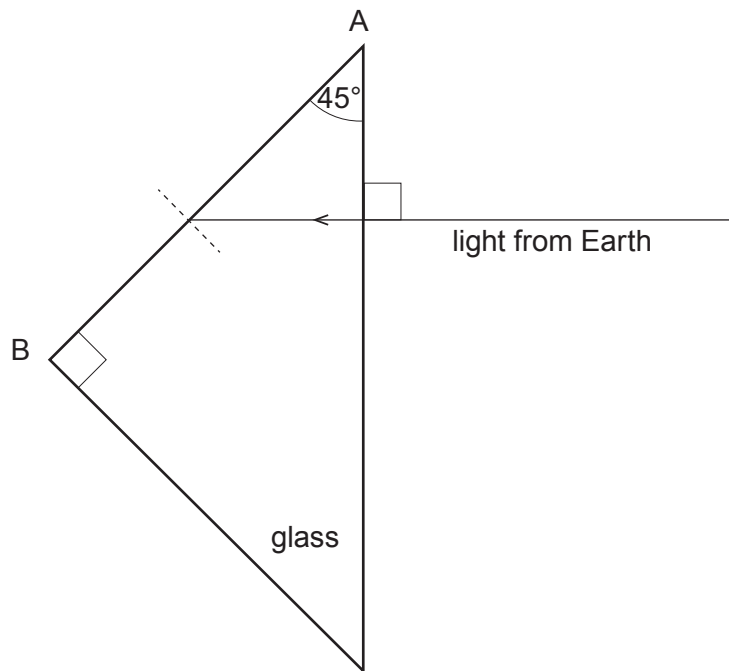
The reflectors have been used to accurately measure the distance from the Moon to Earth and the outcome is that the Moon is found to be getting further away from Earth by about 3.8 cm per year.



Laser light being sent
to the Moon from a
telescope in Germany.

The reflectors were illuminated from Earth by laser light of wavelength 532 nm. The reflectors are made from glass or plastic prisms. The light is sent in pulses separated by short intervals of time from a laser in a telescope on Earth.

The diagram shows a ray of light entering one of the prisms. The critical angle for the glass is 36° .



(i) **Complete the diagram** to show the path followed by the light ray shown. [2]

(ii) Explain why the ray of light would not be reflected back to Earth if it strikes the side AB of the prism at an angle of incidence of 30° . [2]

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(iii) The wavelength of light used for measuring the distance to the Moon is $532 \times 10^{-9} \text{ m}$ and its frequency is $5.639 \times 10^{14} \text{ Hz}$. Use equations from page 2 to calculate the distance between the Earth and the Moon if laser light is directed to the reflector and is received back again on Earth 2.5626 s later. [4]
[All workings must be shown for each stage of your calculation.]

Distance = m

- (iv) State why the laser light needs to be sent to the Moon in pulses separated by at least 2.57 s instead of in a continuous beam. [1]

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END OF PAPER

Examiner
only

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